CSC 201 — Data Structures

Week 8 Lecture Notes

Circular queues reuse empty slots at the front of the array.

A stack overflow occurs when the call stack exceeds its limit.

Recursion solves a problem by calling a smaller version of itself.

Traversal visits every node of a tree exactly once.

Binary search runs in logarithmic time on a sorted array.

Breadth first search uses a queue to explore nodes level by level.

Queues process elements in first come, first served order.

Hash tables map keys to buckets using a hash function.

Insertion at the head of a singly linked list runs in constant time.

A binary tree has at most two children per node.

A linked list stores elements in nodes connected by pointers.

Stacks follow the last in, first out principle.